

CSE400 – Fundamentals of Probability in Computing

Lecture 9: Uniform, Exponential, Laplace, and Gamma Random Variables

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1. Types of Continuous Random Variables

The lecture introduces several types of continuous random variables and their probability distributions:

Uniform Random Variable

Exponential Random Variable

Laplace Random Variable

Gamma Random Variable

Additionally, examples and applications are provided for each distribution.

2. Uniform Random Variable

Definition

A random variable X is uniformly distributed over the interval $[a, b]$ if its probability density function (PDF) is

$$f_X(x) = \begin{cases} \frac{1}{b-a}, & a \leq x < b \\ 0, & \text{elsewhere} \end{cases}$$

The cumulative distribution function (CDF) is

$$F_X(x) = \begin{cases} 0, & x < a \\ \frac{x-a}{b-a}, & a \leq x < b \\ 1, & x \geq b \end{cases}$$

The corresponding PDF and CDF graphs are illustrated in the lecture slides (Figure 3.8).

3. Example 1 – Uniform Random Variable

Problem

The phase of a sinusoid θ is uniformly distributed over $[0, 2\pi]$.

Its PDF is

$$f_{\Theta}(\theta) = \begin{cases} \frac{1}{2\pi}, & 0 \leq \theta < 2\pi \\ 0, & \text{otherwise} \end{cases}$$

Find:

$$Pr(\theta > 3\pi/4)$$

$$Pr(\theta < \pi \mid \theta > 3\pi/4)$$

$$Pr(\cos(\theta) < 1/2)$$

Solution

(a)

$$Pr(\theta > 3\pi/4)$$

For a uniform random variable on $[0, 2\pi]$

$$Pr(a < \theta < b) = \frac{b - a}{2\pi}$$

Thus

$$Pr(\theta > 3\pi/4) = \frac{2\pi - 3\pi/4}{2\pi} = \frac{5}{8}$$

(b) Conditional Probability

$$Pr(A|B) = \frac{Pr(A \cap B)}{Pr(B)}$$

Let

$$A = (\theta < \pi)$$

$$B = (\theta > 3\pi/4)$$

$$Pr(3\pi/4 < \theta < \pi) = \frac{\pi - 3\pi/4}{2\pi} = \frac{1}{8}$$

$$Pr(\theta < \pi \mid \theta > 3\pi/4) = \frac{1/8}{5/8} = \frac{1}{5}$$

(c)

$$Pr(\cos(\theta) < 1/2)$$

$$\cos(\theta) = 1/2 \Rightarrow \theta = \frac{\pi}{3}, \frac{5\pi}{3}$$

Thus

$$\cos(\theta) < 1/2 \quad \text{for} \quad \frac{\pi}{3} < \theta < \frac{5\pi}{3}$$

Therefore

$$Pr(\cos(\theta) < 1/2) = \frac{5\pi/3 - \pi/3}{2\pi} = \frac{4\pi/3}{2\pi} = \frac{2}{3}$$

Applications of Uniform Distribution

Examples include:

Phase of a sinusoidal signal when all phases in $[0, 2\pi]$ are equally likely.

Random number generation between 0 and 1 for simulations.

Arrival time of a user within a known time window assuming no time preference.

4. Exponential Random Variable

Definition

For $b > 0$, the exponential random variable has

PDF

$$f_X(x) = \frac{1}{b} e^{-x/b} u(x)$$

CDF

$$F_X(x) = [1 - e^{-x/b}] u(x)$$

where $u(x)$ is the unit step function.

5. Example 2 – Exponential Random Variable

Problem

Let X be an exponential random variable with

$$f_X(x) = e^{-x} u(x)$$

Find:

$$Pr(3X < 5)$$

$$Pr(3X < y)$$

Let $Y = 3X$. Find $f_Y(y)$.

Solution

Given

$$f_X(x) = e^{-x}u(x)$$

$$F_X(x) = [1 - e^{-x}]u(x)$$

(a)

$$Pr(3X < 5) = Pr(X < 5/3)$$

$$= F_X(5/3)$$

$$= 1 - e^{-5/3}$$

(b)

$$Pr(3X < y) = Pr(X < y/3)$$

$$= F_X(y/3)$$

$$= [1 - e^{-y/3}]u(y)$$

(c)

Let $Y = 3X$

$$F_Y(y) = [1 - e^{-y/3}]u(y)$$

Differentiating

$$f_Y(y) = \frac{d}{dy}F_Y(y)$$

$$f_Y(y) = \frac{1}{3}e^{-y/3}u(y)$$

Thus Y is exponential with parameter

$$\lambda = \frac{1}{3}$$

Applications of Exponential Distribution

Time until radioactive decay of a particle.

Duration of an idle period in a wireless communication channel.

Time until the next earthquake in simplified seismic models.

6. Laplace Random Variable

Definition

PDF

$$f_X(x) = \frac{1}{2b} \exp\left(-\frac{|x|}{b}\right)$$

CDF

$$F_X(x) = \begin{cases} \frac{1}{2} \exp(x/b), & x < 0 \\ 1 - \frac{1}{2} \exp(-x/b), & x \geq 0 \end{cases}$$

Application

The Laplace random variable is used to model the amplitude of speech signals.

7. Example 3 – Laplace Random Variable

Problem

$$f_W(w) = ce^{-2|w|}$$

Find:

c

$Pr(-1 < W < 2)$

$Pr(W > 0 | -1 < W < 2)$

Solution

(a)

$$\int_{-\infty}^{\infty} ce^{-2|w|} dw = 1$$

$$2c \int_0^{\infty} e^{-2w} dw = 1$$

$$2c \left(\frac{1}{2}\right) = 1$$

$$c = 1$$

(b)

$$\begin{aligned}
Pr(-1 < W < 2) &= \int_{-1}^0 e^{2w} dw + \int_0^2 e^{-2w} dw \\
&= \frac{1}{2}(1 - e^{-2}) + \frac{1}{2}(1 - e^{-4}) \\
&= 1 - \frac{1}{2}(e^{-2} + e^{-4})
\end{aligned}$$

(c)

$$\begin{aligned}
Pr(W > 0 | -1 < W < 2) &= \frac{Pr(0 < W < 2)}{Pr(-1 < W < 2)} \\
Pr(0 < W < 2) &= \int_0^2 e^{-2w} dw \\
&= \frac{1}{2}(1 - e^{-4})
\end{aligned}$$

Thus

$$Pr(W > 0 | -1 < W < 2) = \frac{1 - e^{-4}}{2 - (e^{-2} + e^{-4})}$$

8. Gamma Random Variable

PDF

$$f_X(x) = \frac{(x/b)^{c-1} e^{-x/b}}{b\Gamma(c)} u(x)$$

CDF

$$F_X(x) = \frac{\gamma(c, x/b)}{\Gamma(c)} u(x)$$

Definitions

Gamma Function

$$\Gamma(a) = \int_0^\infty e^{-t} t^{a-1} dt$$

Incomplete Gamma Function

$$\gamma(a, \beta) = \int_0^\beta e^{-t} t^{a-1} dt$$

Special Cases

$b = 2, c = 0.5 \rightarrow$ Chi-square random variable

$c = 1 \rightarrow$ Exponential random variable
 with
 $k = c, \theta = b$

9. Example 4 – Gamma Random Variable

Let

$$X \sim \text{Gamma}(\alpha, \lambda)$$

with PDF

$$f_X(x) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}, \quad x \geq 0$$

(a) Expected Value

$$E[X] = \int_0^\infty x f_X(x) dx$$

$$E[X] = \frac{\alpha}{\lambda}$$

(b) Variance

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

$$\text{Var}(X) = \frac{\alpha}{\lambda^2}$$

10. Properties of the Gamma Function

Prove the following

$$\Gamma(n) = (n-1)!$$

for $n = 1, 2, 3, \dots$

$$\Gamma(x+1) = x\Gamma(x)$$

$$\Gamma(1/2) = \sqrt{\pi}$$

11. Applications of the Gamma Distribution

Total service time for multiple independent tasks where each task duration is exponential.

Received signal energy computed from multiple independent signal samples.

Waiting time until the k -th event in a Poisson process.

12. Problem Solving Example – Movie Theater

Suppose arrival time depends on travel time T_r with PDF $f(t)$ and CDF $F(t)$.

Let d be the departure time.

Arrival time

$$Arrival = d + T_r$$

Cost structure

Early arrival

$$cost = c(T - d - T_r)$$

Late arrival

$$cost = k(d + T_r - T)$$

Expected cost

$$E[C(d)] = \int_0^{T-d} c(T - d - t)f(t)dt + \int_{T-d}^{\infty} k(t + d - T)f(t)dt$$

Optimal Departure Time

$$-cF(T - d) + k[1 - F(T - d)] = 0$$

Thus

$$F(T - d^*) = \frac{k}{c + k}$$

$$d^* = T - F^{-1}\left(\frac{k}{c + k}\right)$$

Interpretation

Depart such that the probability of arriving early equals

$$\frac{k}{c + k}$$

13. CDF Analysis Example

Given

$$F_X(x) = \begin{cases} 0, & x < 0 \\ x^2, & 0 \leq x \leq 1 \\ 1, & x > 1 \end{cases}$$

PDF

$$f_X(x) = \frac{d}{dx}F_X(x) = 2x, \quad 0 \leq x \leq 1$$

Mean

$$\mu_X = \int_0^1 x(2x)dx$$

$$\mu_X = \frac{2}{3}$$

Variance

$$\sigma_X^2 = \int_0^1 x^2(2x)dx - \mu_X^2$$

$$= \frac{1}{2} - \frac{4}{9}$$

$$\sigma_X^2 = \frac{1}{18}$$

Skewness

$$c_s = \frac{E[(X - \mu)^3]}{\sigma^3} = -\frac{2\sqrt{2}}{5}$$

Kurtosis

$$c_k = \frac{E[(X - \mu)^4]}{\sigma^4} = \frac{12}{5}$$

Conclusion

Distribution is left-skewed

Platykurtic (lighter tails than Gaussian)